

MTH2021 Mathematical Methods 2

Self-study sheet 3

1. Evaluate $(-1)^{i/\pi}$.
2. Using the definitions of complex powers and the complex logarithm, find the distinct values of $i^{\frac{1}{3}}$.
3. Verify that the real and imaginary parts of $f(z) = \sin z$ solve Laplace's equation in two dimensions.
4. (a) Show that the power series for the exponential function

$$\exp z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$$

is absolutely convergent for all z .

- (b) Starting from the above power series, show that

i. $\exp z \exp w = \exp(z + w)$

ii. $\exp(iz) = \cos z + i \sin z$ where

$$\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} \quad \text{and} \quad \sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$$

5. Show that the series

$$C = \sum_{k=0}^{\infty} p^k \cos k\theta \quad \text{and} \quad S = \sum_{k=0}^{\infty} p^k \sin k\theta$$

for p, θ real with $|p| < 1$ are absolutely convergent and evaluate them.

6. Integrate

$$f(z) = \frac{z^*}{1 + |z|^2}$$

around the closed path from $z = -1$ to $z = 1$ along the real axis, and then along the arc $z = \exp(it)$ from $t = 0$ to $t = \pi$.
